Program 39: Design a web page by applying CSS Display and Positioning property.

Objective: - To design a web page layout using CSS *display* and *position* properties to control element alignment, visibility, and placement effectively.

Code: -

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Display & Position</title>
<style>
.container{
  border:2px solid black;
  padding:20px;
  width:300px;
}
.box{
  width:100px;
  height:100px;
  background:lightblue;
  display:inline-block;
  position:relative;
  left:50px;
  text-align:center;
  line-height:100px;
}
</style>
</head>
<body>
<h3 align="center">Name:- Aditya Gangwar</h3>
<h4 align="center">Roll No:- 2401330100023</h4>
```

<hr>

BOX

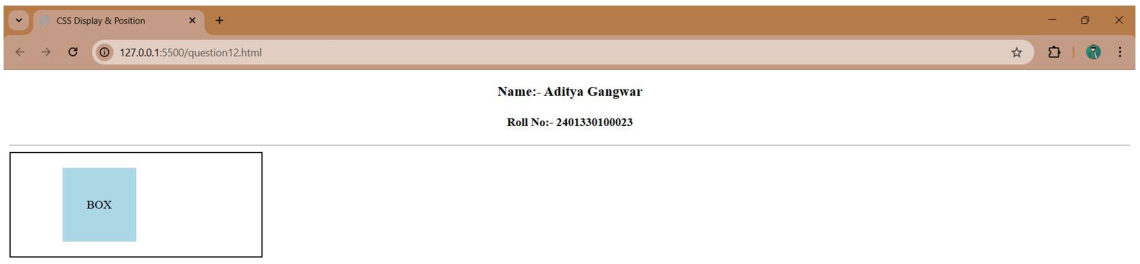
</body>

</html>

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Display & Position</title>
<style>
.container{
  border:2px solid black;
  padding:20px;
  width:300px;
}
.box{
  width:100px;
  height:100px;
  background: lightblue;
  display:inline-block;
  position:relative;
  left:50px;
  text-align:center;
  line-height:100px;
}
</style>
</head>
<body>
<h3 align="center">Name:- Aditya Gangwar</h3>
<h4 align="center">Roll No:- 2401330100023</h4>
<hr>
<div class="container">
  <div class="box">BOX</div>
</div>
</body>
</html>

```

Output: -



Program 40: Design a web page by applying CSS Display and Positioning property.

Objective: - To create a web page layout using CSS display and positioning properties to control the arrangement and placement of elements on the page.

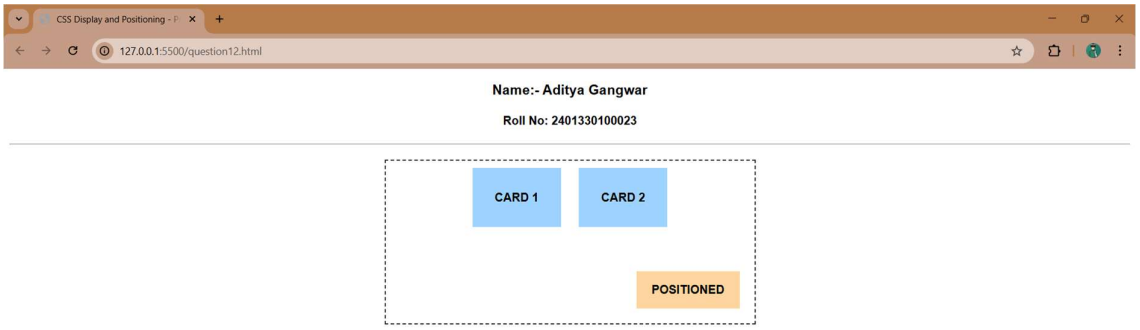
Code: -

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Display and Positioning - Program 40</title>
<style>
body{
  font-family: Arial, sans-serif;
  text-align: center;
}
.wrapper{
  width: 500px;
  height: 220px;
  border: 2px dashed #333;
  margin: 20px auto;
  position: relative;
}
.card{
  width: 120px;
  height: 80px;
  background-color: #9fd3ff;
  display: inline-block;
  margin: 10px;
  line-height: 80px;
  font-weight: bold;
}
```

```
.badge{
  width: 140px;
  height: 50px;
  background-color: #ffd59f;
  position: absolute;
  bottom: 20px;
  right: 20px;
  line-height: 50px;
  font-weight: bold;
}
</style>
</head>
<body>
<h3>Name:- Aditya Gangwar</h3>
<h4>Roll No: 2401330100023</h4>
<hr>
<div class="wrapper">
  <div class="card">CARD 1</div>
  <div class="card">CARD 2</div>
  <div class="badge">POSITIONED</div>
</div>
</body>
</html>
```

```html
<!DOCTYPE html>
<html>
<head>
<title>CSS Display and Positioning - Program 40</title>
<style>
body{
  font-family: Arial, sans-serif;
  text-align: center;
}
.wrapper{
  width: 500px;
  height: 220px;
  border: 2px dashed #333;
  margin: 20px auto;
  position: relative;
}
.card{
  width: 120px;
  height: 80px;
  background-color: #9fd3ff;
  display: inline-block;
  margin: 10px;
  line-height: 80px;
  font-weight: bold;
}
.badge{
  width: 140px;
  height: 50px;
  background-color: #ffd59f;
  position: absolute;
  bottom: 20px;
  right: 20px;
  line-height: 50px;
  font-weight: bold;
}
</style>
</head>
<body>
<h3>Name:- Aditya Gangwar</h3>
<h4>Roll No: 2401330100023</h4>
<hr>
<div class="wrapper">
  <div class="card">CARD 1</div>
  <div class="card">CARD 2</div>
  <div class="badge">POSITIONED</div>
</div>
</body>
</html>
```

Output: -



Program 41: Design a web page by applying CSS pseudo classes.

Objective: - To design a web page using CSS pseudo-classes to enhance interactivity and visual effects based on user actions and element states.

Code: -

```
<!DOCTYPE html>
<html>
<head>
<title>CSS Pseudo Classes - Program 41</title>
<style>
body{
  font-family: Arial, sans-serif;
  text-align: center;
}
/* Pseudo class :hover */
a:hover{
  color: red;
  font-size: 20px;
}
/* Pseudo class :focus */
input:focus{
  background-color: lightyellow;
  border: 2px solid green;
}
/* Pseudo class :active */
button:active{
  background-color: orange;
}
```

```
/* Pseudo class :visited */
a:visited{
  color: purple;
}
</style>
</head>
<body>
<h3>Name:- Aditya Gangwar</h3>
<h4>Roll No: 2401330100023</h4>
<hr>
<p>
  <a href="#">Hover / Visit Me</a>
</p>
<p>
  <input type="text" placeholder="Focus here">
</p>
<p>
  <button>Click Button</button>
</p>
</body>
</html>
```

```html
<!DOCTYPE html>
<html>
<head>
<title>CSS Pseudo Classes - Program 41</title>
<style>
body{
  font-family: Arial, sans-serif;
  text-align: center;
}
/* Pseudo class :hover */
a:hover{
  color: red;
  font-size: 20px;
}
/* Pseudo class :focus */
input:focus{
  background-color: lightyellow;
  border: 2px solid green;
}
/* Pseudo class :active */
button:active{
  background-color: orange;
}

/* Pseudo class :visited */
a:visited{
  color: purple;
}
</style>
</head>
<body>
<h3>Name:- Aditya Gangwar</h3>
<h4>Roll No: 2401330100023</h4>
<hr>
<p>
  <a href="#">Hover / Visit Me</a>
</p>
<p>
  <input type="text" placeholder="Focus here">
</p>
<p>
  <button>Click Button</button>
</p>
</body>
</html>
```

Output: -



Program 42: Creating a Java Script code to implement all data types.

Objective: -To implement and demonstrate all JavaScript data types using a simple JavaScript program.

Code: -

```
<!DOCTYPE html>
<html>
<head>
<title>JavaScript Data Types - Program 42</title>
<style>
body{
  font-family: Arial, sans-serif;
  text-align: center;
}
.box{
  display: inline-block;
  border: 2px solid black;
  padding: 15px 25px;
  margin-top: 20px;
  text-align: left;
}
</style>
</head>
<body>

<h3>Name:- Aditya Gangwar</h3>
<h4>Roll No:- 2401330100023</h4>
<hr>
<div class="box">
<script>
let num = 100;
let str = "JavaScript";
```

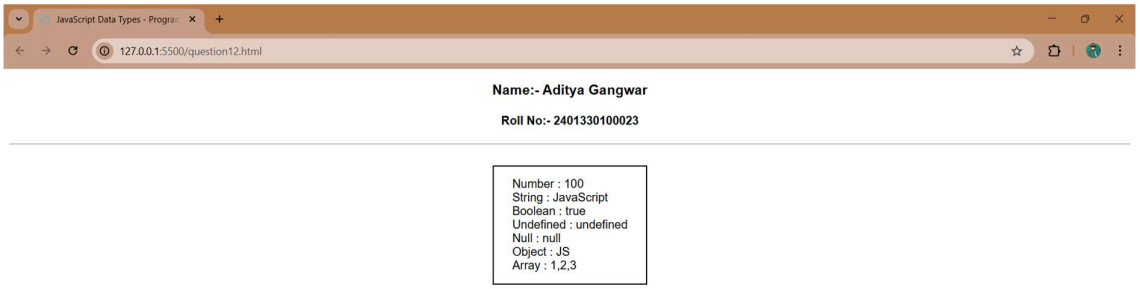
```
let bool = true;
let undef;
let nul = null;
let obj = {language:"JS"};
let arr = [1,2,3];
document.write("Number : " + num + "<br>");
document.write("String : " + str + "<br>");
document.write("Boolean : " + bool + "<br>");
document.write("Undefined : " + undef + "<br>");
document.write("Null : " + nul + "<br>");
document.write("Object : " + obj.language + "<br>");
document.write("Array : " + arr);
</script>
</div>
</body>
</html>
```

```html
<!DOCTYPE html>
<html>
<head>
<title>JavaScript Data Types - Program 42</title>
<style>
body{
  font-family: Arial, sans-serif;
  text-align: center;
}
.box{
  display: inline-block;
  border: 2px solid black;
  padding: 15px 25px;
  margin-top: 20px;
  text-align: left;
}
</style>
</head>
<body>

<h3>Name:- Aditya Gangwar</h3>
<h4>Roll No:- 2401330100023</h4>
<hr>
<div class="box">
<script>
let num = 100;
let str = "JavaScript";
let bool = true;
let undef;
let nul = null;
let obj = {language:"JS"};
let arr = [1,2,3];
document.write("Number : " + num + "<br>");
document.write("String : " + str + "<br>");
document.write("Boolean : " + bool + "<br>");
document.write("Undefined : " + undef + "<br>");
document.write("Null : " + nul + "<br>");
document.write("Object : " + obj.language + "<br>");
document.write("Array : " + arr);
</script>
</div>
</body>
</html>

```

Output: -



Program 43: Design a basic structure of Bootstrap Grid system.

Objective: -To design a basic web page layout using the Bootstrap Grid system for creating responsive rows and columns.

Code: -

<!DOCTYPE html>

<html>

<head>

<link rel="stylesheet"

href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">

</head>

<body class="text-center">

<h3>Name:- Aditya Gangwar</h3>

<h4>Roll No:- 2401330100023</h4>

<hr>

<button class="btn btn-primary">Bootstrap Button</button>

This is Alert Box

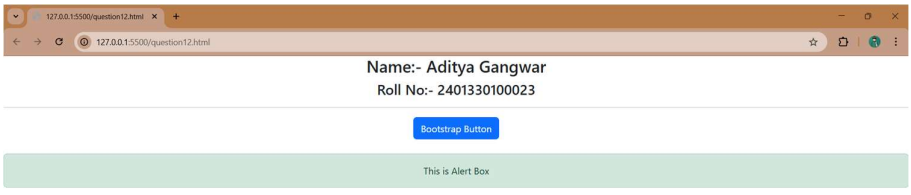
</body>

</html>

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
</head>
<body class="text-center">
<h3>Name:- Aditya Gangwar</h3>
<h4>Roll No:- 2401330100023</h4>
<hr>
<button class="btn btn-primary">Bootstrap Button</button>
<br><br>
<div class="alert alert-success">This is Alert Box</div>
</body>
</html>

```

Output: -



Program 44: Design All Bootstrap Components with example.

Objective: -To design and demonstrate commonly used Bootstrap components with examples to understand their structure, styling, and functionality.

Code: -

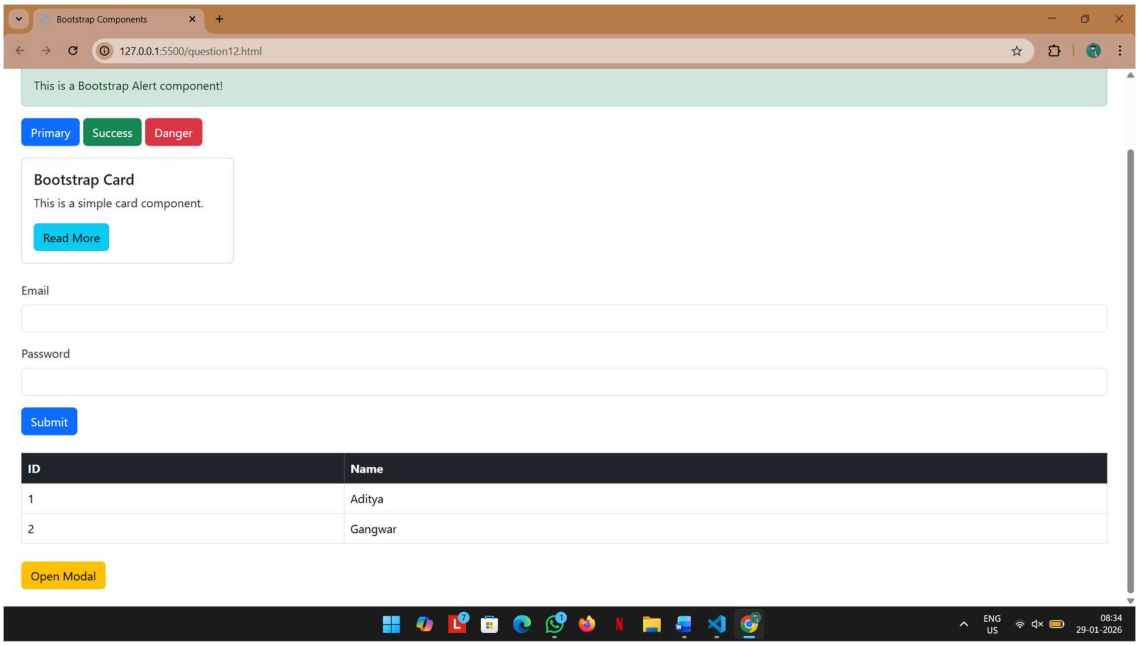
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Bootstrap Components</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body class="p-4">
  <nav class="navbar navbar-expand-lg navbar-dark bg-dark mb-4">
    <div class="container-fluid">
      <a class="navbar-brand" href="#">Bootstrap Demo</a>
      <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#menu">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div id="menu" class="collapse navbar-collapse">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item"><a class="nav-link" href="#">Home</a></li>
          <li class="nav-item"><a class="nav-link" href="#">About</a></li>
        </ul>
      </div>
    </div>
  </nav>
  <div class="alert alert-success">
    This is a Bootstrap Alert component!
  </div>
  <div class="mb-3">
```

```
    <button class="btn btn-primary">Primary</button>
    <button class="btn btn-success">Success</button>
    <button class="btn btn-danger">Danger</button>
</div>
<div class="card mb-4" style="width: 18rem;">
    <div class="card-body">
        <h5 class="card-title">Bootstrap Card</h5>
        <p class="card-text">This is a simple card component.</p>
        <a href="#" class="btn btn-info">Read More</a>
    </div>
</div>
<form class="mb-4">
    <div class="mb-3">
        <label class="form-label">Email</label>
        <input type="email" class="form-control">
    </div>
    <div class="mb-3">
        <label class="form-label">Password</label>
        <input type="password" class="form-control">
    </div>
    <button class="btn btn-primary">Submit</button>
</form>
<table class="table table-bordered mb-4">
    <thead class="table-dark">
        <tr>
            <th>ID</th>
            <th>Name</th>
        </tr>
    </thead>
    <tbody>
```

```
        <tr>
          <td>1</td>
          <td>Aditya</td>
        </tr>
        <tr>
          <td>2</td>
          <td>Gangwar</td>
        </tr>
      </tbody>
    </table>
    <button class="btn btn-warning" data-bs-toggle="modal" data-bs-target="#myModal">
      Open Modal
    </button>
    <div class="modal fade" id="myModal">
      <div class="modal-dialog">
        <div class="modal-content">
          <div class="modal-header">
            <h5 class="modal-title">Bootstrap Modal</h5>
            <button class="btn-close" data-bs-dismiss="modal"></button>
          </div>
          <div class="modal-body">
            This is a modal example.
          </div>
        </div>
      </div>
    </div>
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Bootstrap Components</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body class="p-4">
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark mb-4">
        <div class="container-fluid">
            <a class="navbar-brand" href="#">Bootstrap Demo</a>
            <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#menu">
                <span class="navbar-toggler-icon"></span>
            </button>
            <div id="menu" class="collapse navbar-collapse">
                <ul class="navbar-nav ms-auto">
                    <li class="nav-item"><a class="nav-link" href="#">Home</a></li>
                    <li class="nav-item"><a class="nav-link" href="#">About</a></li>
                </ul>
            </div>
        </div>
    </nav>
    <div class="alert alert-success">
        This is a Bootstrap Alert component!
    </div>
    <div class="mb-3">
        <button class="btn btn-primary">Primary</button>
        <button class="btn btn-success">Success</button>
        <button class="btn btn-danger">Danger</button>
    </div>
    <div class="card mb-4" style="width: 18rem;">
        <div class="card-body">
            <h5 class="card-title">Bootstrap Card</h5>
            <p class="card-text">This is a simple card component.</p>
            <a href="#" class="btn btn-info">Read More</a>
        </div>
    </div>
    <form class="mb-4">
        <div class="mb-3">
            <label class="form-label">Email</label>
            <input type="email" class="form-control">
        </div>
        <div class="mb-3">
            <label class="form-label">Password</label>
            <input type="password" class="form-control">
        </div>
        <button class="btn btn-primary">Submit</button>
    </form>
    <table class="table table-bordered mb-4">
        <thead class="table-dark">
            <tr>
                <th>ID</th>
                <th>Name</th>
            </tr>
        </thead>
        <tbody>
            <tr>
                <td>1</td>
                <td>Aditya</td>
            </tr>
            <tr>
                <td>2</td>
                <td>Gangwar</td>
            </tr>
        </tbody>
    </table>
    <button class="btn btn-warning" data-bs-toggle="modal" data-bs-target="#myModal">
        Open Modal
    </button>
    <div class="modal fade" id="myModal">
        <div class="modal-dialog">
            <div class="modal-content">
                <div class="modal-header">
                    <h5 class="modal-title">Bootstrap Modal</h5>
                    <button class="btn-close" data-bs-dismiss="modal"></button>
                </div>
                <div class="modal-body">
                    This is a modal example.
                </div>
            </div>
        </div>
    </div>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Output: -

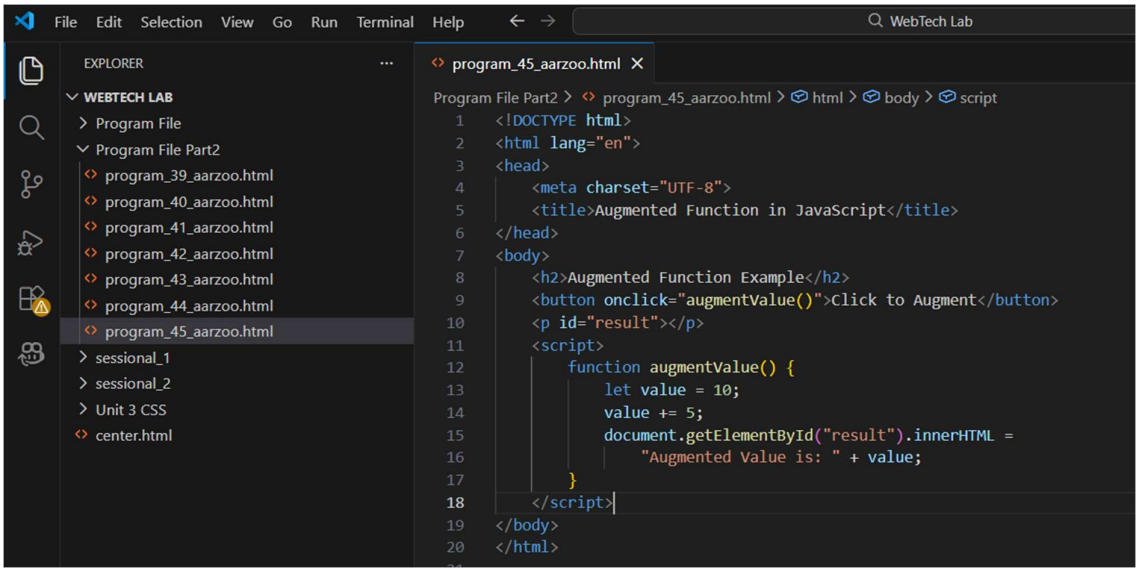

Bootstrap Components
127.0.0.1:5500/question12.html
This is a Bootstrap Alert component!
Primary
Success
Danger
Bootstrap Card
This is a simple card component.
Read More
Email
Password
Submit
ID
Name
1
Aditya
2
Gangwar
Open Modal
ENG
US
08:34
29-01-2026

Program 45: Implementing a program in Java script to implement augmented function.

Objective: -To implement a JavaScript program that uses an augmented function to modify and return an updated value.

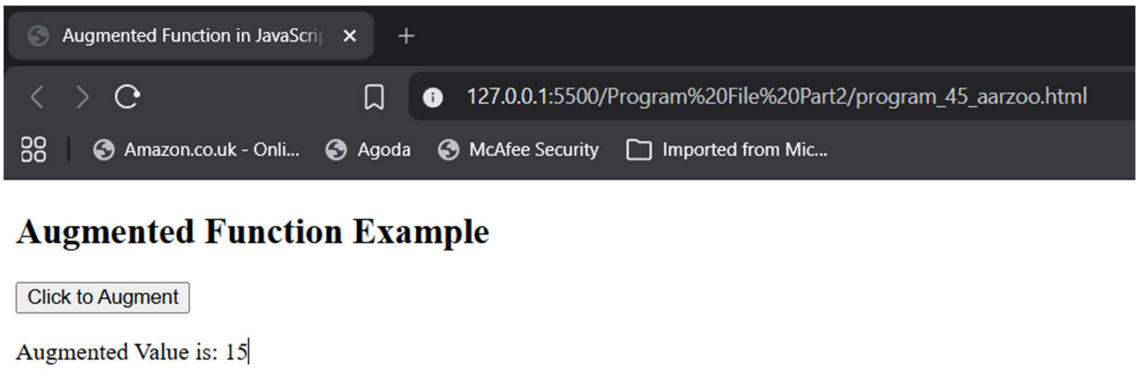
Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Augmented Function in JavaScript</title>
</head>
<body>
    <h2>Augmented Function Example</h2>
    <button onclick="augmentValue()">Click to Augment</button>
    <p id="result"></p>
    <script>
        function augmentValue() {
            let value = 10;
            value += 5;
            document.getElementById("result").innerHTML =
                "Augmented Value is: " + value;
        }
    </script>
</body>
</html>
```


File Edit Selection View Go Run Terminal Help
WebTech Lab
EXPLORER
WEBTECH LAB
Program File
Program File Part2
program_39_aarzoo.html
program_40_aarzoo.html
program_41_aarzoo.html
program_42_aarzoo.html
program_43_aarzoo.html
program_44_aarzoo.html
program_45_aarzoo.html
sessional_1
sessional_2
Unit 3 CSS
center.html
program_45_aarzoo.html
Program File Part2 > program_45_aarzoo.html > html > body > script
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Augmented Function in JavaScript</title>
</head>
<body>
<h2>Augmented Function Example</h2>
<button onclick="augmentValue()">Click to Augment</button>

<script>
function augmentValue() {
let value = 10;
value += 5;
document.getElementById("result").innerHTML =
"Augmented Value is: " + value;
}
</script>
</body>
</html>

Output: -


Augmented Function in JavaScri
127.0.0.1:5500/Program%20File%20Part2/program_45_aarzoo.html
Amazon.co.uk - Onli...
Agoda
McAfee Security
Imported from Mic...
Augmented Function Example
Click to Augment
Augmented Value is: 15

Program 46: Implementing a program to implement calculator application as real time.

Objective: -To implement a real-time calculator application using HTML and JavaScript for performing basic arithmetic operations.

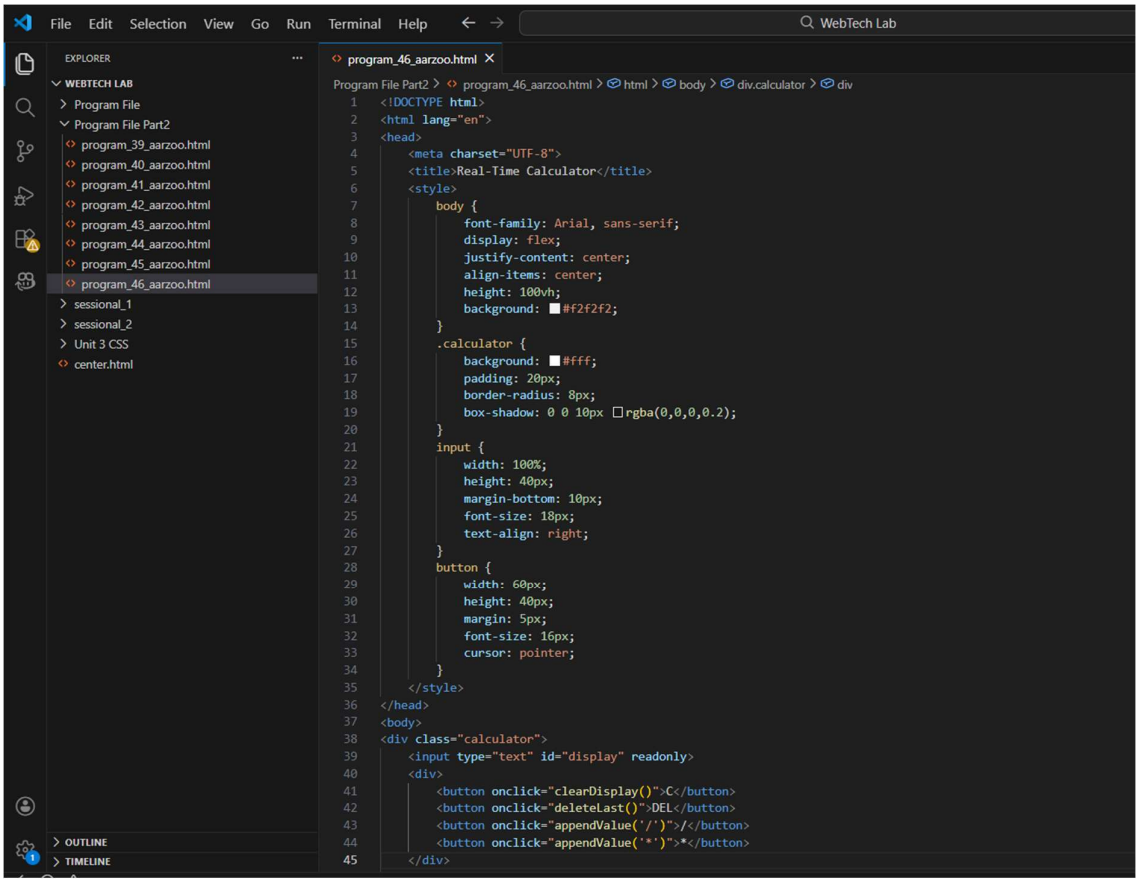
Code: -

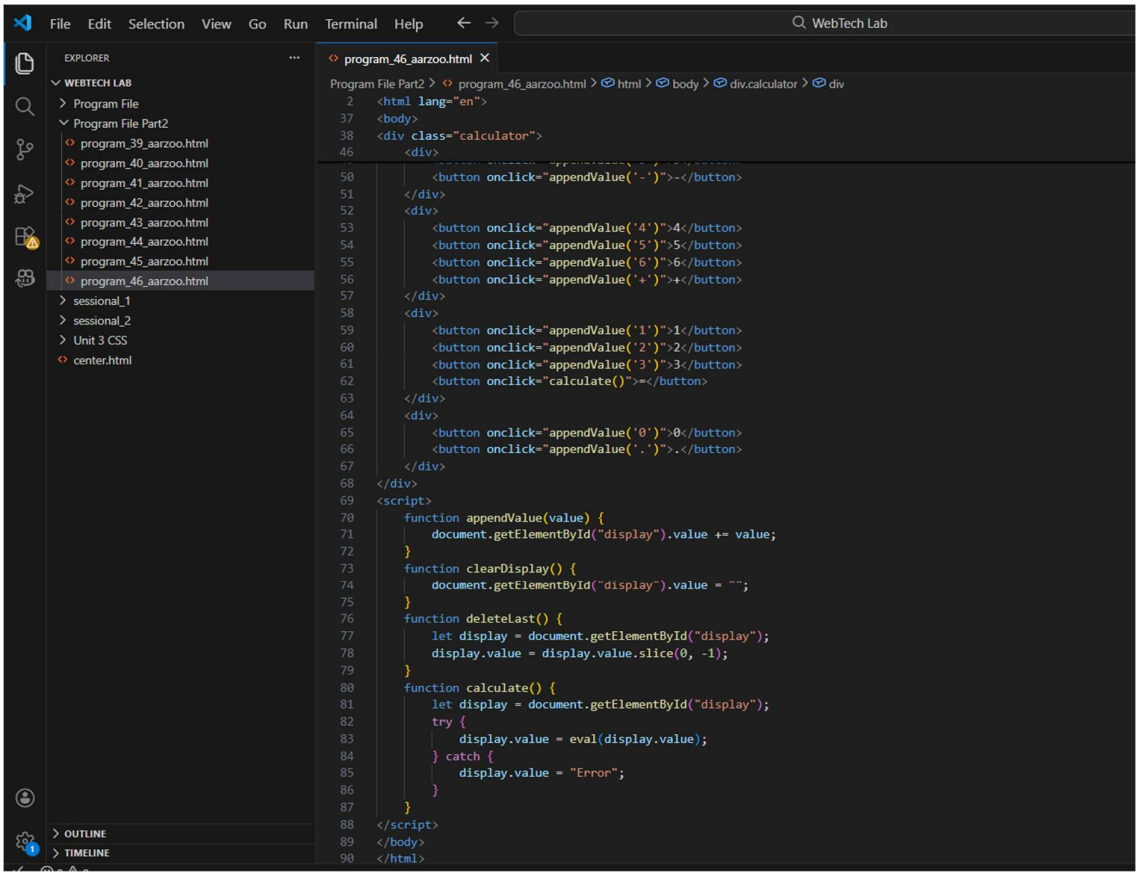
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Real-Time Calculator</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            background: #f2f2f2;
        }
        .calculator {
            background: #fff;
            padding: 20px;
            border-radius: 8px;
            box-shadow: 0 0 10px rgba(0,0,0,0.2);
        }
        input {
            width: 100%;
            height: 40px;
            margin-bottom: 10px;
            font-size: 18px;
            text-align: right;
```

```
        }
        button {
            width: 60px;
            height: 40px;
            margin: 5px;
            font-size: 16px;
            cursor: pointer;
        }
    </style>
</head>
<body>
<div class="calculator">
    <input type="text" id="display" readonly>
    <div>
        <button onclick="clearDisplay()">C</button>
        <button onclick="deleteLast()">DEL</button>
        <button onclick="appendValue('/')">/</button>
        <button onclick="appendValue('*')">*</button>
    </div>
    <div>
        <button onclick="appendValue('7')">7</button>
        <button onclick="appendValue('8')">8</button>
        <button onclick="appendValue('9')">9</button>
        <button onclick="appendValue('-')">-</button>
    </div>
    <div>
        <button onclick="appendValue('4')">4</button>
        <button onclick="appendValue('5')">5</button>
        <button onclick="appendValue('6')">6</button>
        <button onclick="appendValue('+')">+</button>
```

```
    </div>
    <div>
        <button onclick="appendValue('1')">1</button>
        <button onclick="appendValue('2')">2</button>
        <button onclick="appendValue('3')">3</button>
        <button onclick="calculate()">=</button>
    </div>
    <div>
        <button onclick="appendValue('0')">0</button>
        <button onclick="appendValue('.')">.</button>
    </div>
</div>
<script>
    function appendValue(value) {
        document.getElementById("display").value += value;
    }
    function clearDisplay() {
        document.getElementById("display").value = "";
    }
    function deleteLast() {
        let display = document.getElementById("display");
        display.value = display.value.slice(0, -1);
    }
    function calculate() {
        let display = document.getElementById("display");
        try {
            display.value = eval(display.value);
        } catch {
            display.value = "Error";
        }
```

```
    }
</script>
</body>
</html>
```




File Edit Selection View Go Run Terminal Help
WebTech Lab
EXPLORER
WEBTECH LAB
Program File
Program File Part2
program_39_aarzoo.html
program_40_aarzoo.html
program_41_aarzoo.html
program_42_aarzoo.html
program_43_aarzoo.html
program_44_aarzoo.html
program_45_aarzoo.html
program_46_aarzoo.html
sessional_1
sessional_2
Unit 3 CSS
center.html
OUTLINE
TIMELINE
program_46_aarzoo.html
Program File Part2 > program_46_aarzoo.html > html > body > div.calculator > div
<html lang="en">
<body>

<button onclick="appendValue('-')">-</button>

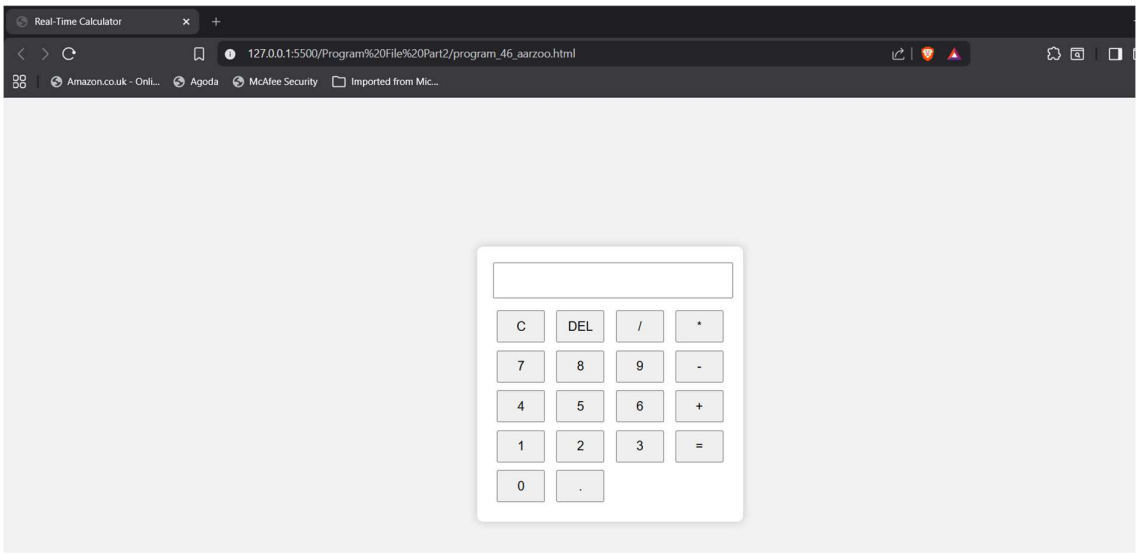
<button onclick="appendValue('4')">4</button>
<button onclick="appendValue('5')">5</button>
<button onclick="appendValue('6')">6</button>
<button onclick="appendValue('+')">+</button>

<button onclick="appendValue('1')">1</button>
<button onclick="appendValue('2')">2</button>
<button onclick="appendValue('3')">3</button>
<button onclick="calculate()">=</button>

<button onclick="appendValue('0')">0</button>
<button onclick="appendValue('.')">.</button>

<script>
function appendValue(value) {
document.getElementById("display").value += value;
}
function clearDisplay() {
document.getElementById("display").value = "";
}
function deleteLast() {
let display = document.getElementById("display");
display.value = display.value.slice(0, -1);
}
function calculate() {
let display = document.getElementById("display");
try {
display.value = eval(display.value);
} catch {
display.value = "Error";
}
}
</script>
</body>
</html>

Output: -


Real-Time Calculator
127.0.0.1:5500/Program%20File%20Part2/program_46_aarzoo.html
Amazon.co.uk - Onli...
Agoda
McAfee Security
Imported from Mic...
C
DEL
/
*
7
8
9
-
4
5
6
+
1
2
3
=
0
.

Program 47: Design a HTML form validation using Java Script.

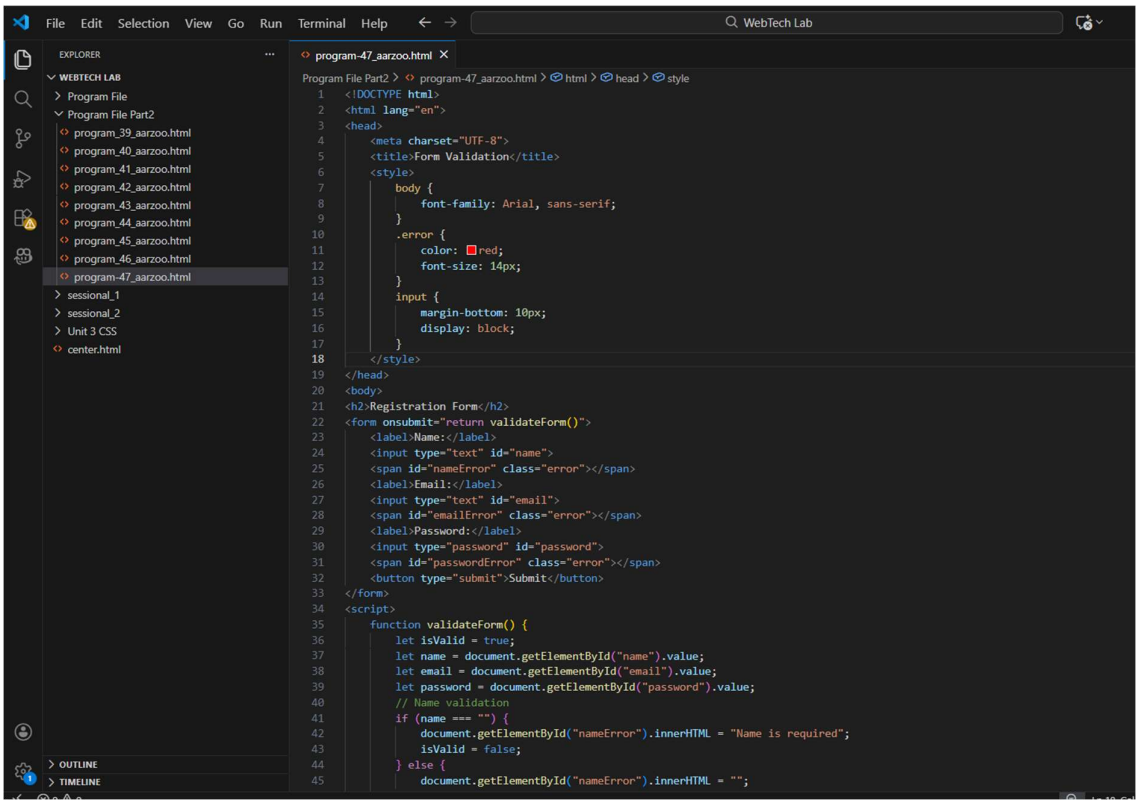
Objective: - To design an HTML form and apply JavaScript validation to ensure correct and complete user input before form submission.

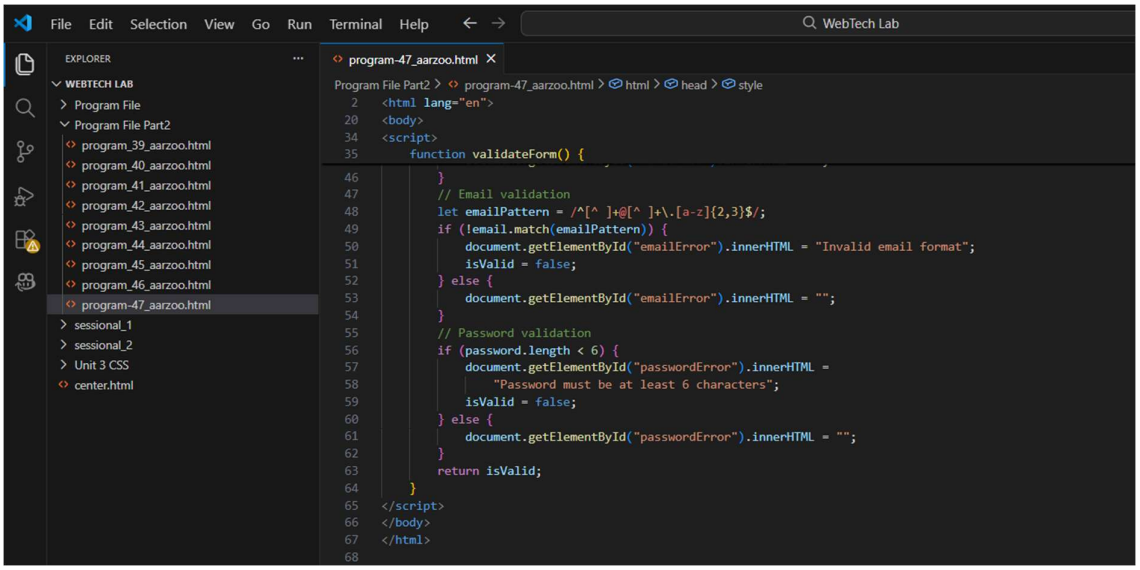
Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Form Validation</title>
    <style>
        body {
            font-family: Arial, sans-serif;
        }
        .error {
            color: red;
            font-size: 14px;
        }
        input {
            margin-bottom: 10px;
            display: block;
        }
    </style>
</head>
<body>
<h2>Registration Form</h2>
<form onsubmit="return validateForm()">
    <label>Name:</label>
    <input type="text" id="name">
    <span id="nameError" class="error"></span>
    <label>Email:</label>
    <input type="text" id="email">
```

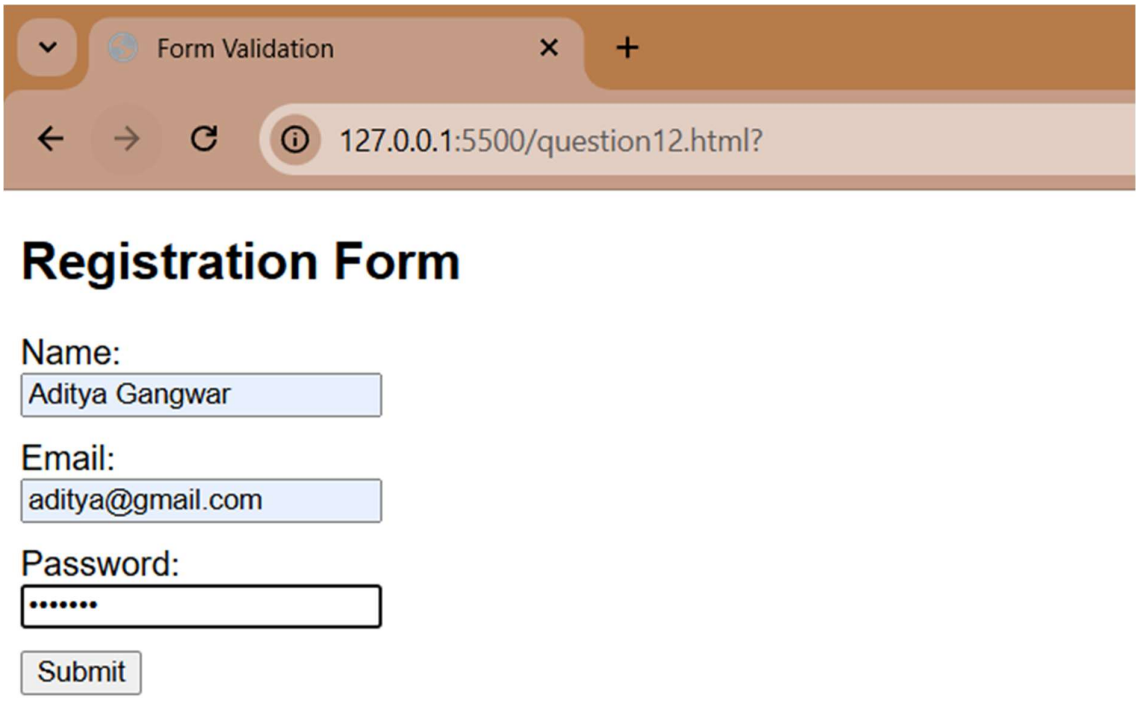
```
    <span id="emailError" class="error"></span>
    <label>Password:</label>
    <input type="password" id="password">
    <span id="passwordError" class="error"></span>
    <button type="submit">Submit</button>
</form>
<script>
    function validateForm() {
        let isValid = true;
        let name = document.getElementById("name").value;
        let email = document.getElementById("email").value;
        let password = document.getElementById("password").value;
        // Name validation
        if (name === "") {
            document.getElementById("nameError").innerHTML = "Name is required";
            isValid = false;
        } else {
            document.getElementById("nameError").innerHTML = "";
        }
        // Email validation
        let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
        if (!email.match(emailPattern)) {
            document.getElementById("emailError").innerHTML = "Invalid email format";
            isValid = false;
        } else {
            document.getElementById("emailError").innerHTML = "";
        }
        // Password validation
        if (password.length < 6) {
            document.getElementById("passwordError").innerHTML =
```

```
            "Password must be at least 6 characters";
            isValid = false;
        } else {
            document.getElementById("passwordError").innerHTML = "";
        }
        return isValid;
    }
</script>
</body>
</html>
```




File Edit Selection View Go Run Terminal Help
WebTech Lab
EXPLORER
WEBTECH LAB
Program File
Program File Part2
program_39_aarzoo.html
program_40_aarzoo.html
program_41_aarzoo.html
program_42_aarzoo.html
program_43_aarzoo.html
program_44_aarzoo.html
program_45_aarzoo.html
program_46_aarzoo.html
program-47_aarzoo.html
sessional_1
sessional_2
Unit 3 CSS
center.html
program-47_aarzoo.html
Program File Part2 > program-47_aarzoo.html > html > head > style
2 <html lang="en">
20 <body>
34 <script>
35 function validateForm() {
46 }
47 // Email validation
48 let emailPattern = /^[^]+@[^]+\.[a-z]{2,3}$/;
49 if (!email.match(emailPattern)) {
50 document.getElementById("emailError").innerHTML = "Invalid email format";
51 isValid = false;
52 } else {
53 document.getElementById("emailError").innerHTML = "";
54 }
55 // Password validation
56 if (password.length < 6) {
57 document.getElementById("passwordError").innerHTML =
58 "Password must be at least 6 characters";
59 isValid = false;
60 } else {
61 document.getElementById("passwordError").innerHTML = "";
62 }
63 return isValid;
64 }
65 </script>
66 </body>
67 </html>
68

Output:-


Form Validation
127.0.0.1:5500/question12.html?
Registration Form
Name:
Aditya Gangwar
Email:
aditya@gmail.com
Password:
Submit

Program 48: Write a program to implement Arrow function with default argument in ES6.

Objective: - To implement an ES6 arrow function using a default argument to handle missing parameters efficiently.

Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Arrow Function with Default Argument</title>
</head>
<body>
<h2>Arrow Function Example</h2>
<p id="output"></p>

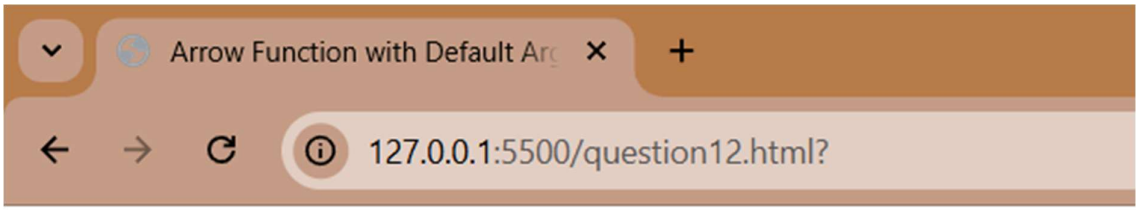
<script>
    // Arrow function with default argument
    const greet = (name = "Guest") => {
        return "Welcome, " + name + "!";
    };
    // Function calls
    document.getElementById("output").innerHTML =
        greet("Aditya") + "<br>" + greet();
</script>
</body>
</html>
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Arrow Function with Default Argument</title>
</head>
<body>
<h2>Arrow Function Example</h2>
<p id="output"></p>

<script>
    // Arrow function with default argument
    const greet = (name = "Guest") => {
        return "Welcome, " + name + "!";
    };
    // Function calls
    document.getElementById("output").innerHTML =
        greet("Aditya") + "<br>" + greet();
</script>
</body>
</html>

```

Output: -



Arrow Function Example

Welcome, Aditya!
Welcome, Guest!

Program 49: Implementing a program in ES6 to implement Template string concepts.

Objective: -To implement ES6 template strings for dynamic string creation and variable embedding in JavaScript.

Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>ES6 Template Strings</title>
</head>
<body>
<h2>Template String Example</h2>
<p id="result"></p>
<script>
    let name = "Aditya";
    let course = "Web Technology";
    let year = 2026;

    let message = `Hello ${name}!
You are enrolled in ${course} for the year ${year}.`;
    document.getElementById("result").innerText = message;
</script>
</body>
</html>
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>ES6 Template Strings</title>
</head>
<body>
<h2>Template String Example</h2>
<p id="result"></p>
<script>
    let name = "Aditya";
    let course = "Web Technology";
    let year = 2026;

    let message = `Hello ${name}!
You are enrolled in ${course} for the year ${year}.`;
    document.getElementById("result").innerText = message;
</script>
</body>
</html>

```

Output: -



Template String Example

Hello Aditya!
You are enrolled in Web Technology for the year 2026.

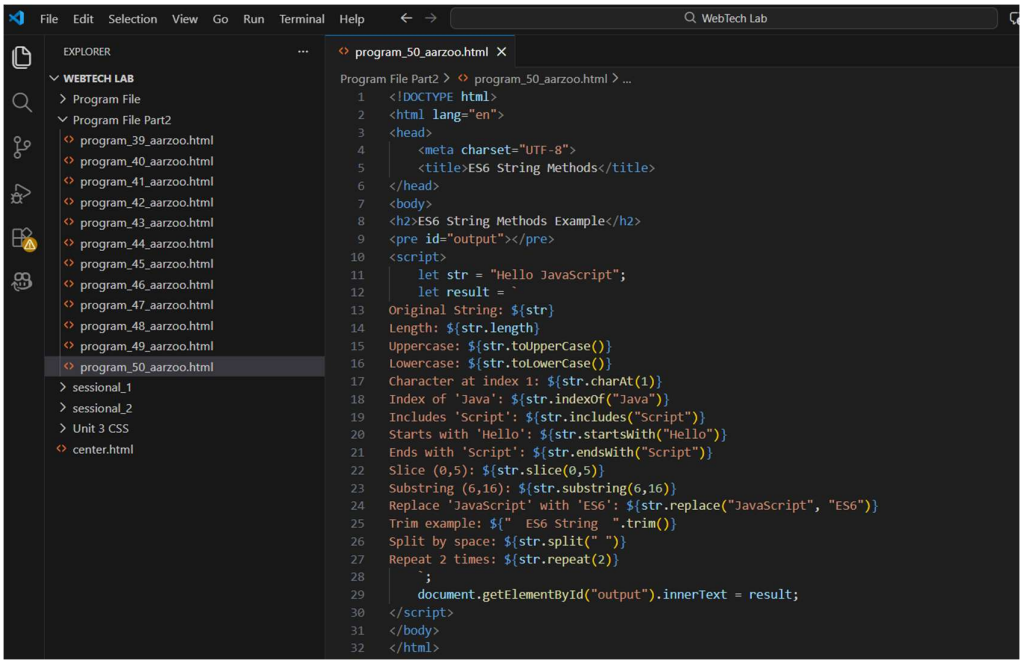
Program 50: Implementing a program in ES6 to implement all string methods.

Objective: -To implement an ES6 program demonstrating commonly used JavaScript string methods for string manipulation.

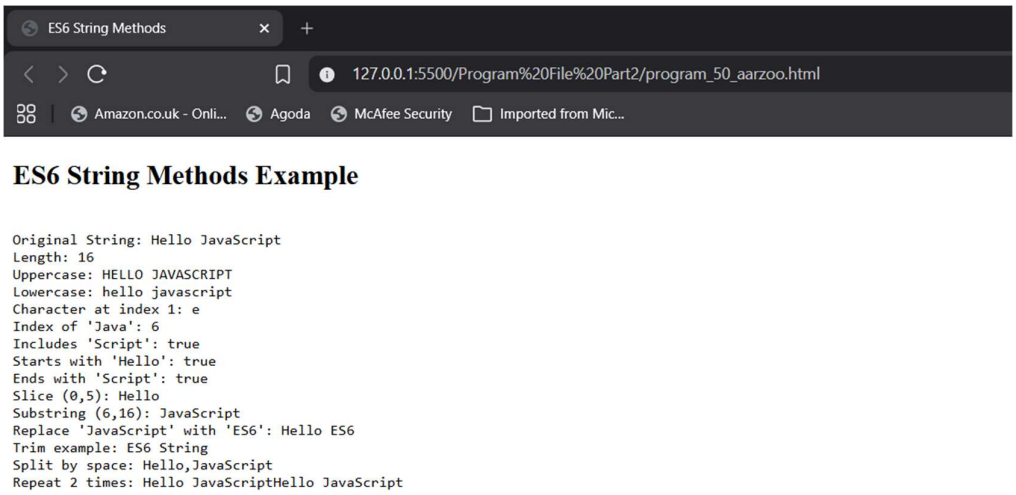
Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>ES6 String Methods</title>
</head>
<body>
<h2>ES6 String Methods Example</h2>
<pre id="output"></pre>
<script>
    let str = "Hello JavaScript";
    let result = `
Original String: ${str}
Length: ${str.length}
Uppercase: ${str.toUpperCase()}
Lowercase: ${str.toLowerCase()}
Character at index 1: ${str.charAt(1)}
Index of 'Java': ${str.indexOf("Java")}
Includes 'Script': ${str.includes("Script")}
Starts with 'Hello': ${str.startsWith("Hello")}
Ends with 'Script': ${str.endsWith("Script")}
Slice (0,5): ${str.slice(0,5)}
Substring (6,16): ${str.substring(6,16)}
Replace 'JavaScript' with 'ES6': ${str.replace("JavaScript", "ES6")}
Trim example: ${"  ES6 String  ".trim()}
Split by space: ${str.split(" ")}
```

```
Repeat 2 times: ${str.repeat(2)}
    `;
    document.getElementById("output").innerText = result;
</script>
</body>
</html>
```



Output: -



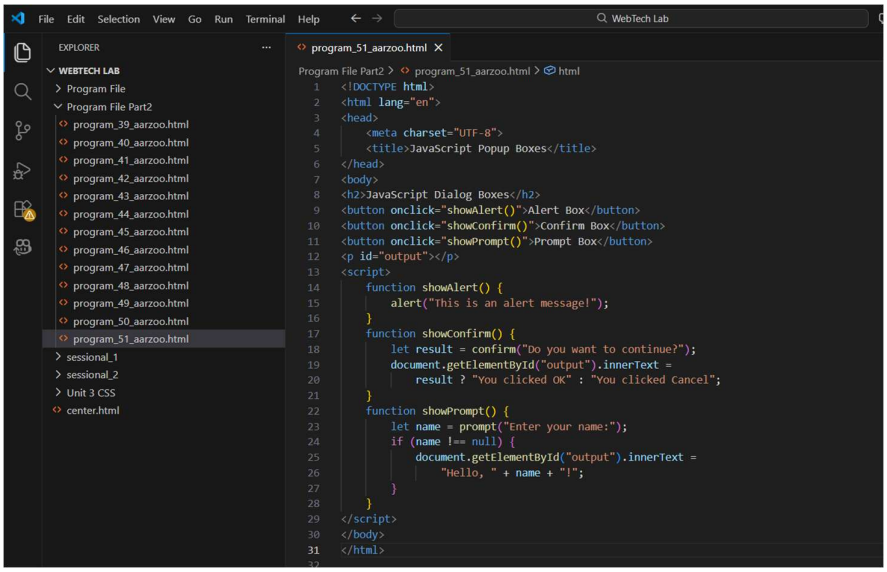
Program 51: Creating a Java Script program to implement Dialog, Confirm and Message Popup Boxes.

Objective: -To create a JavaScript program that demonstrates dialog boxes using alert, confirm, and prompt pop-up messages.

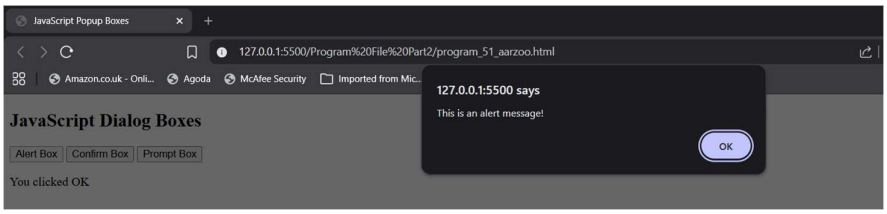
Code: -

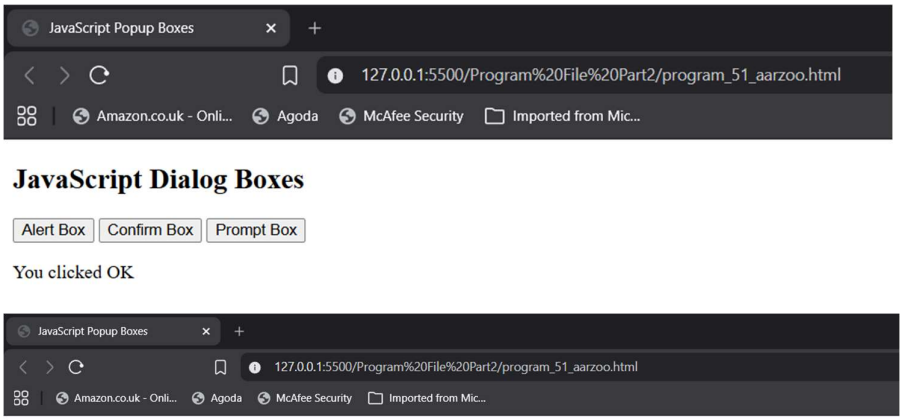
```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>JavaScript Popup Boxes</title>
</head>
<body>
<h2>JavaScript Dialog Boxes</h2>
<button onclick="showAlert()">Alert Box</button>
<button onclick="showConfirm()">Confirm Box</button>
<button onclick="showPrompt()">Prompt Box</button>
<p id="output"></p>
<script>
   function showAlert() {
      alert("This is an alert message!");
   }
   function showConfirm() {
      let result = confirm("Do you want to continue?");
      document.getElementById("output").innerText =
         result ? "You clicked OK" : "You clicked Cancel";
   }
   function showPrompt() {
      let name = prompt("Enter your name:");
      if (name !== null) {
         document.getElementById("output").innerText =
            "Hello, " + name + "!";
```

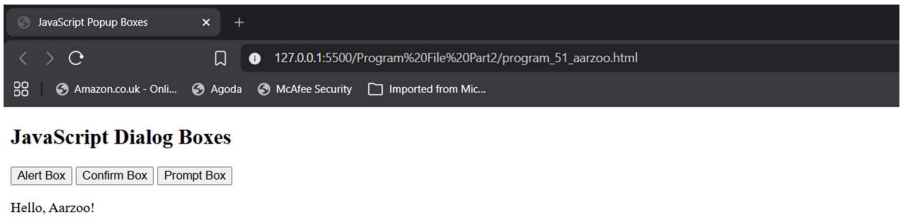
```
        }
    }
</script>
</body>
</html>
```



Output: -







Program 52: Implementing a Java Script program to implement onClick and onSubmit event.

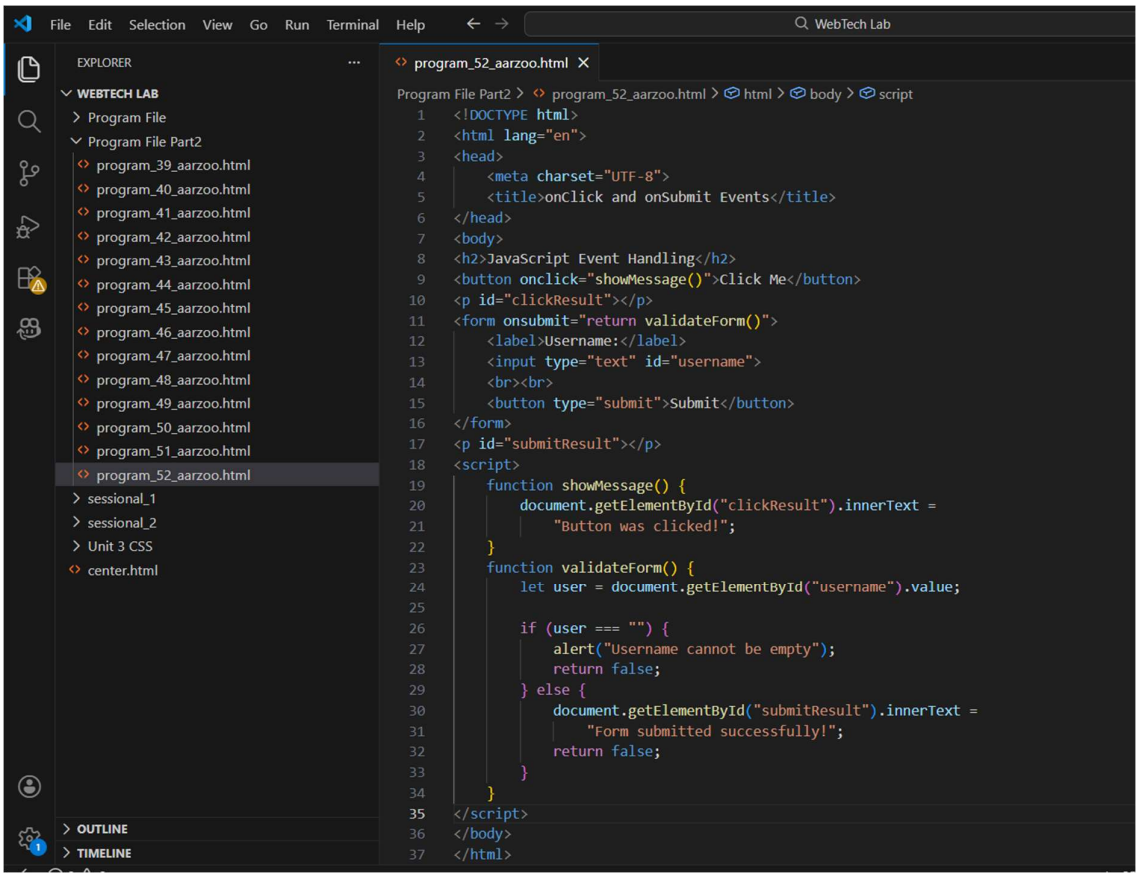
Objective: -To implement JavaScript onClick and onSubmit events to handle user interactions on a web page.

Code: -

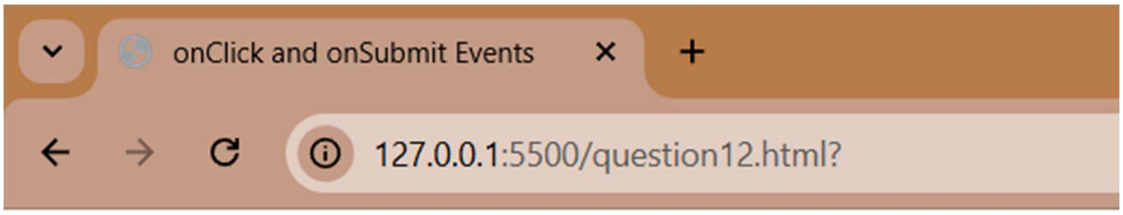
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>onClick and onSubmit Events</title>
</head>
<body>
<h2>JavaScript Event Handling</h2>
<button onclick="showMessage()">Click Me</button>
<p id="clickResult"></p>
<form onsubmit="return validateForm()">
    <label>Username:</label>
    <input type="text" id="username">
    <br><br>
    <button type="submit">Submit</button>
</form>
<p id="submitResult"></p>
<script>
    function showMessage() {
        document.getElementById("clickResult").innerText =
            "Button was clicked!";
    }
    function validateForm() {
        let user = document.getElementById("username").value;

        if (user === "") {
```

```
            alert("Username cannot be empty");
            return false;
        } else {
            document.getElementById("submitResult").innerText =
                "Form submitted successfully!";
            return false;
        }
    }
</script>
</body>
</html>
```



Output: -



JavaScript Event Handling

Click Me

Button was clicked!

Username: 0241CSE191

Submit

Form submitted successfully!

Program 53: Creating a java script code to implement ‘let’ keyword.

Objective: -To implement the JavaScript let keyword to declare block-scoped variables.

Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>JavaScript let Keyword</title>
</head>
<body>

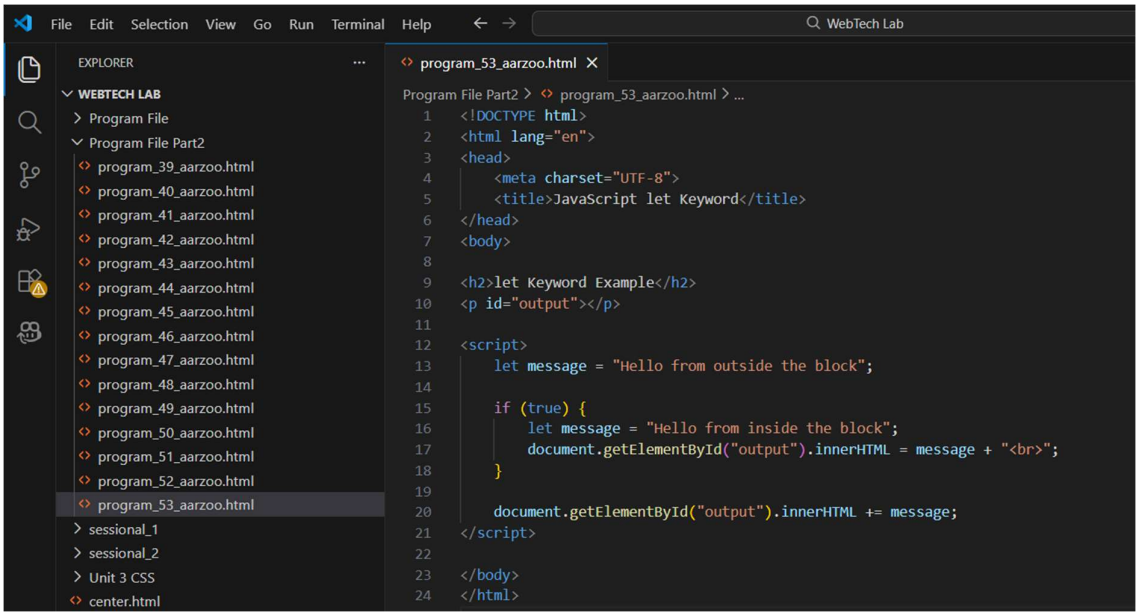
<h2>let Keyword Example</h2>
<p id="output"></p>

<script>
   let message = "Hello from outside the block";

   if (true) {
      let message = "Hello from inside the block";
      document.getElementById("output").innerHTML = message + "<br>";
   }

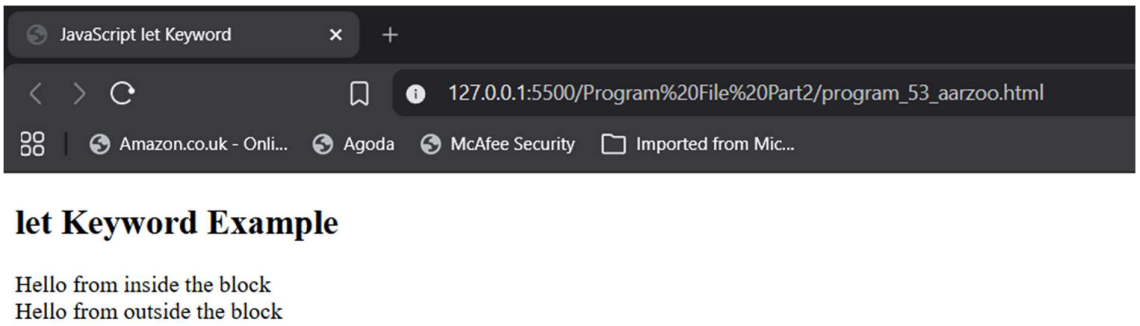
   document.getElementById("output").innerHTML += message;
</script>

</body>
</html>
```


File Edit Selection View Go Run Terminal Help
WebTech Lab
EXPLORER
WEBTECH LAB
Program File
Program File Part2
program_39_aarzoo.html
program_40_aarzoo.html
program_41_aarzoo.html
program_42_aarzoo.html
program_43_aarzoo.html
program_44_aarzoo.html
program_45_aarzoo.html
program_46_aarzoo.html
program_47_aarzoo.html
program_48_aarzoo.html
program_49_aarzoo.html
program_50_aarzoo.html
program_51_aarzoo.html
program_52_aarzoo.html
program_53_aarzoo.html
sessional_1
sessional_2
Unit 3 CSS
center.html
program_53_aarzoo.html
Program File Part2 > program_53_aarzoo.html > ...
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>JavaScript let Keyword</title>
</head>
<body>
<h2>let Keyword Example</h2>

<script>
let message = "Hello from outside the block";
if (true) {
let message = "Hello from inside the block";
document.getElementById("output").innerHTML = message + "
";
}
document.getElementById("output").innerHTML += message;
</script>
</body>
</html>

Output: -


JavaScript let Keyword
127.0.0.1:5500/Program%20File%20Part2/program_53_aarzoo.html
Amazon.co.uk - Onli...
Agoda
McAfee Security
Imported from Mic...
let Keyword Example
Hello from inside the block
Hello from outside the block

Program 54: Creating a java script code to implement ‘const’ keyword.

Objective: -To implement the JavaScript const keyword for declaring variables with constant values.

Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>JavaScript const Keyword</title>
</head>
<body>

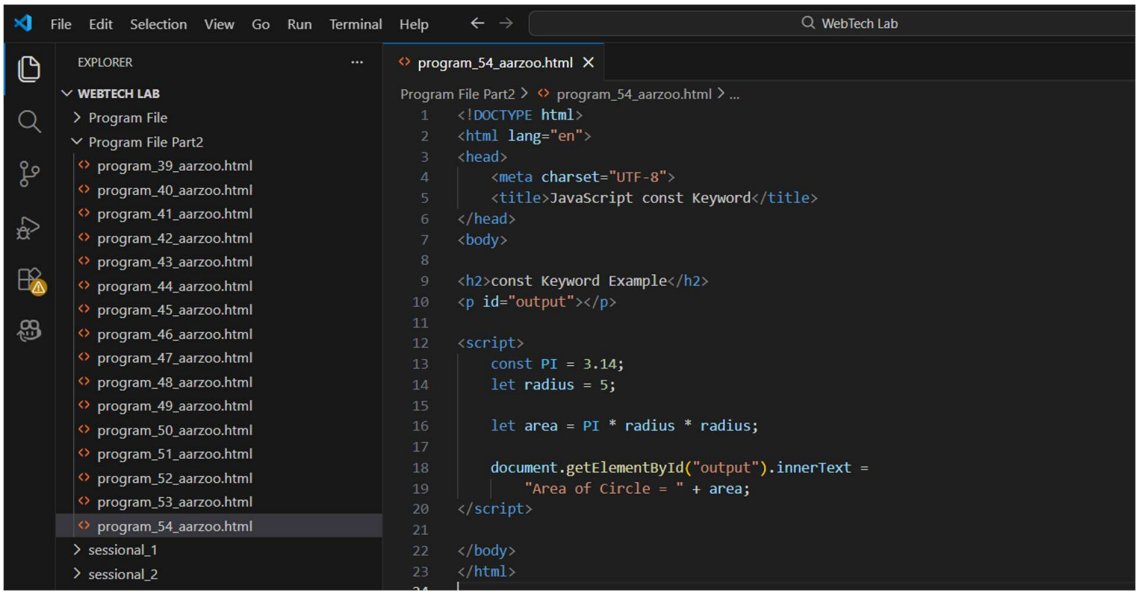
<h2>const Keyword Example</h2>
<p id="output"></p>

<script>
    const PI = 3.14;
    let radius = 5;

    let area = PI * radius * radius;

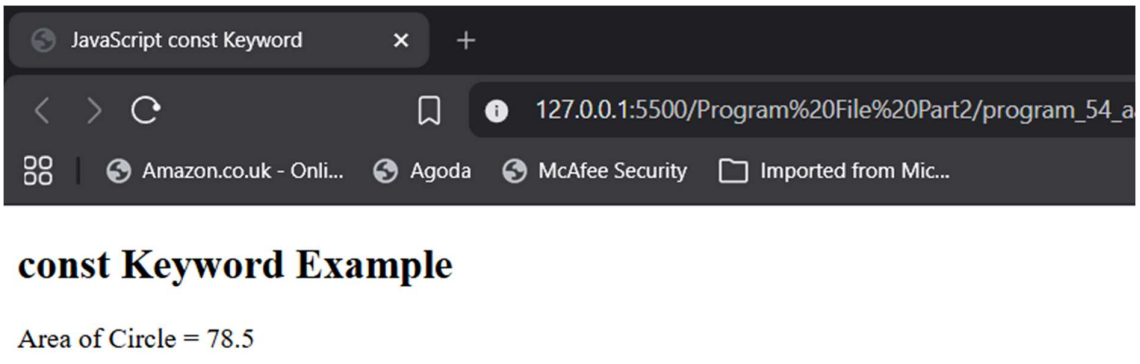
    document.getElementById("output").innerText =
        "Area of Circle = " + area;
</script>

</body>
</html>
```


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WebTech Lab
EXPLORER
WEBTECH LAB
Program File
Program File Part2
program_39_aarzoo.html
program_40_aarzoo.html
program_41_aarzoo.html
program_42_aarzoo.html
program_43_aarzoo.html
program_44_aarzoo.html
program_45_aarzoo.html
program_46_aarzoo.html
program_47_aarzoo.html
program_48_aarzoo.html
program_49_aarzoo.html
program_50_aarzoo.html
program_51_aarzoo.html
program_52_aarzoo.html
program_53_aarzoo.html
program_54_aarzoo.html
sessional_1
sessional_2
program_54_aarzoo.html
Program File Part2 > program_54_aarzoo.html > ...
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>JavaScript const Keyword</title>
</head>
<body>
<h2>const Keyword Example</h2>

<script>
const PI = 3.14;
let radius = 5;
let area = PI * radius * radius;
document.getElementById("output").innerText =
"Area of Circle = " + area;
</script>
</body>
</html>

Output: -


JavaScript const Keyword
127.0.0.1:5500/Program%20File%20Part2/program_54_a
Amazon.co.uk - Onli...
Agoda
McAfee Security
Imported from Mic...
const Keyword Example
Area of Circle = 78.5

Program 55: Implementing a program to implement call back functions in ES6.

Objective: - To implement ES6 callback functions to execute a function after another function completes.

Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>ES6 Callback Function</title>
</head>
<body>

<h2>Callback Function Example</h2>
<p id="output"></p>

<script>
    // Callback function
    const displayMessage = (message) => {
        document.getElementById("output").innerText = message;
    };

    // Function that accepts callback
    const processData = (name, callback) => {
        let result = `Hello ${name}, callback executed successfully!`;
        callback(result);
    };

    // Function call
    processData("Aditya", displayMessage);
</script>
```

</body>

</html>

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>ES6 Callback Function</title>
</head>
<body>

<h2>Callback Function Example</h2>
<p id="output"></p>

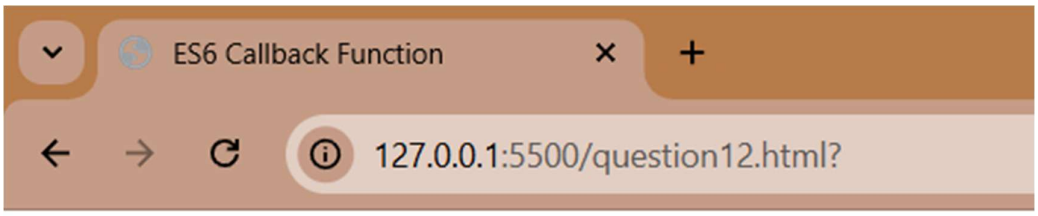
<script>
    // Callback function
    const displayMessage = (message) => {
        document.getElementById("output").innerText = message;
    };

    // Function that accepts callback
    const processData = (name, callback) => {
        let result = `Hello ${name}, callback executed successfully!`;
        callback(result);
    };

    // Function call
    processData("Aditya", displayMessage);
</script>

</body>
</html>
```

Output: -



Callback Function Example

Hello Aditya, callback executed successfully!

Program 56: Implementing a program for de-structuring of an array in ES6.

Objective: -To implement ES6 array destructuring for extracting values from an array into separate variables.

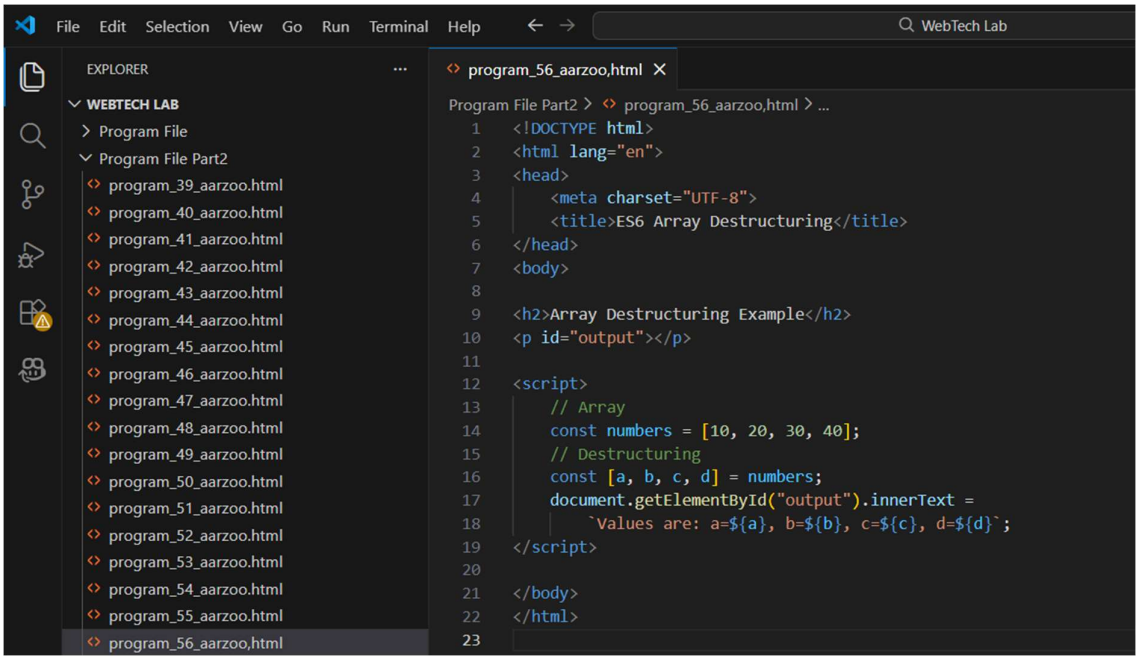
Code: -

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>ES6 Array Destructuring</title>
</head>
<body>

<h2>Array Destructuring Example</h2>
<p id="output"></p>

<script>
    // Array
    const numbers = [10, 20, 30, 40];
    // Destructuring
    const [a, b, c, d] = numbers;
    document.getElementById("output").innerText =
        `Values are: a=${a}, b=${b}, c=${c}, d=${d}`;
</script>

</body>
</html>
```


File Edit Selection View Go Run Terminal Help
WebTech Lab
EXPLORER
WEBTECH LAB
Program File
Program File Part2
program_39_aarzoo.html
program_40_aarzoo.html
program_41_aarzoo.html
program_42_aarzoo.html
program_43_aarzoo.html
program_44_aarzoo.html
program_45_aarzoo.html
program_46_aarzoo.html
program_47_aarzoo.html
program_48_aarzoo.html
program_49_aarzoo.html
program_50_aarzoo.html
program_51_aarzoo.html
program_52_aarzoo.html
program_53_aarzoo.html
program_54_aarzoo.html
program_55_aarzoo.html
program_56_aarzoo.html
program_56_aarzoo,html
Program File Part2 > program_56_aarzoo,html > ...
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>ES6 Array Destructuring</title>
</head>
<body>
<h2>Array Destructuring Example</h2>

<script>
// Array
const numbers = [10, 20, 30, 40];
// Destructuring
const [a, b, c, d] = numbers;
document.getElementById("output").innerText =
`Values are: a=${a}, b=${b}, c=${c}, d=${d}`;
</script>
</body>
</html>

Output: -


ES6 Array Destructuring
127.0.0.1:5500/Program%20File
Amazon.co.uk - Onli...
Agoda
McAfee Security
Imported fr
Array Destructuring Example
Values are: a=10, b=20, c=30, d=40